Matteo Falasca

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PD Dr. Sylke Blumstengel
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PhD application "Hybrid-plasmonic platform with tuneable light–matter coupling in a 2D parameter space" job ID DR/153/26

Dear Dr. Blumstengel,

I am completing my MSc in theoretical physics at the University of Trieste, where I expect to graduate in December 2026, and I am applying for the doctoral position in your group. My training is in light–matter interactions, many-body physics and numerical computation. I am applying for this position because I want to continue my studies in a field where theory can be simulated and where my knowledge can be useful to real state-of-the-art experiments and implementations.

Light–matter interaction is at the centre of my Master's thesis, "Graviton Scattering and Induced Decoherence", supervised by Prof. Angelo Bassi. I study how a bosonic field that scatters off matter changes the dynamics of the matter itself. I quantized the field coupled to matter and derived the scattering cross sections, which generalize the light–matter case and include velocity corrections I could not find in the literature. The work gave me a solid command of quantum field theory and second quantization, and a clear picture of how the coupling to a field defines the behaviour of matter: the same physics your platform makes tuneable.

Your project aims to demonstrate the potential of plasmonic platforms, partly through numerical finite element simulations, and simulation and data analysis are where my experience is strongest. In my Bachelor's thesis I analysed eROSITA data and built a neural-network emulator for a parameter space too large to sample directly, using Latin hypercube sampling. Sampling a parameter space efficiently would be a huge resource for a platform tuneable in a 2D parameter space like yours. For the noisy-circuit work of Grossi et al. (arXiv:2410.01893) I wrote a faster algorithm that computes information flows and that was later used in their research, and in a computational physics course I acquired experience from Quantum Monte Carlo to simulating systems with emerging collective behaviours. I code in Python and Fortran and can use software tools for data processing and analysis.

I built my background in condensed matter physics and optics through my choice of courses, starting with Atoms, Molecules and Photons with Prof. Francesco Scazza and Quantum Many-Body Systems with Prof. Federico Becca, neither of which was mandatory for my curriculum, together with several courses on Statistical Mechanics and Quantum Field Theory. My interest in these topics grew in 2025, when I was selected for the three-week ICTP School on Quantum Dynamics of Matter, Light and Information, and when I attended the DPG Bad Honnef school on quantum metrology, both closely related with light–matter interactions.

I have not yet used optical spectroscopy in the lab, but I have met it on the theory side. Among my study projects, the one for Atoms, Molecules and Photons consisted of presenting research results at a domestic conference on the creation of noise-resistant entangled states of light by the addition of a single delocalized photon, and it led me to study homodyne detection and how quantum states of light are measured. This is a part of physics I would like to get into: performing measurements and building theory next to verifiable experiments fascinates me, I am looking for the chance to do it. Personally, I would feel very comfortable presenting results in international journals and at conferences: I have been used to public speaking since I was young, and I stay calm in front of an audience.

On a personal level, I am an outgoing person and I like working with others, with whom I build healthy working environments. I tutored first-year Bachelor students, and I have refereed football since 2017, now in the top regional division, which taught me to decide under pressure, to work in a team and to explain my choices.

Prof. Angelo Bassi and Prof. Francesco Scazza have agreed to write reference letters for me and can send them directly to you. Thank you for your time and consideration.

Yours sincerely,

Matteo Falasca

Code: Some of my projects